



INFINITE SERIES

D'ALEMBERT's RATIO TEST:

A positive term series $\sum u_n$

i) Converges, if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lambda < 1$,

ii) Diverges, if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lambda > 1$,

iii) Test fails, if $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lambda = 1$.



$$1) \sum \frac{n!}{n^n}$$

$$u_n = \frac{n!}{n^n} \quad u_{n+1} = \frac{(n+1)!}{(n+1)^{(n+1)}}$$

$$\frac{u_{n+1}}{u_n} = \frac{(n+1)!}{(n+1)^{(n+1)}} \cdot \frac{n^n}{n!}$$

$$= \left(\frac{n}{n+1}\right)^n$$

$$= \frac{1}{(1+1/n)^n}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{1}{(1+1/n)^n}$$

$$= \frac{1}{e} < 1.$$

By ratio test, the series is convergent.



$$2) \sum \frac{n^2}{2^n}$$

$$u_n = \frac{n^2}{2^n} \quad u_{n+1} = \frac{(n+1)^2}{2^{n+1}}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{2} \left(1 + \frac{1}{n}\right)^2$$

$$= \frac{1}{2} < 1$$

By ratio test, the series is convergent.



$$3) \frac{1}{2\sqrt{1}} + \frac{x^2}{3\sqrt{2}} + \frac{x^4}{4\sqrt{3}} + \frac{x^6}{5\sqrt{4}} + \dots \infty$$

$$u_n = \frac{x^{2n-2}}{(n+1)\sqrt{n}} \quad \& \quad u_{n+1} = \frac{x^{2n}}{(n+2)\sqrt{(n+1)}}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{x^{2n}}{(n+2)\sqrt{(n+1)}} \cdot \frac{(n+1)\sqrt{n}}{x^{2n-2}}$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)}{(n+2)} \cdot \left(\frac{n}{n+1} \right)^{1/2} x^2$$

$$= \lim_{n \rightarrow \infty} \frac{1+1/n}{1+2/n} \cdot \frac{1}{\sqrt{(1+1/n)}} x^2$$

$$= x^2$$

Hence $\sum u_n$ converges if $x^2 < 1$ & *diverges if* $x^2 > 1$



If $x^2 = 1$, then $u_n = \frac{1}{(n+1)\sqrt{n}}$

$$= \frac{1}{n^{3/2}} \cdot \frac{1}{(1+1/n)}$$

Taking $v_n = \frac{1}{n^{3/2}}$, we get

$$\lim_{n \rightarrow \infty} \frac{u_n}{v_n} = \lim_{n \rightarrow \infty} \frac{1}{1+1/n} = 1 \neq 0, \text{ a finite quantity.}$$

Both $\sum u_n$ & $\sum v_n$ converge or diverge together.

But $\sum v_n = \sum \frac{1}{n^{3/2}}$ is a convergent series.

Hence the given series converges if $x^2 \leq 1$ & diverges if $x^2 > 1$.



$$4) \frac{2}{5}x + \frac{6}{9}x^2 + \frac{14}{17}x^3 + \dots + \frac{2^n - 2}{2^{n+1}}x^{n-1} + \dots$$

$$\frac{u_{n+1}}{u_n} = \frac{2^{n+1} - 2}{2^{n+1} + 1} x^n \cdot \frac{2^{n+1}}{2^n - 2} \frac{1}{x^{n-1}}$$

$$= \frac{2^{-2/2^n}}{2^{+1/2^n}} \cdot \frac{1^{+1/2^n}}{1^{-2/2^n}} x$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \frac{2-0}{2+0} \cdot \frac{1+0}{1-0} x = x$$

Thus by ratio test, $\sum u_n$ converges for $x < 1$ & and diverges for $x > 1$. But it fails for $x = 1$



$$\text{When } x = 1, \lim_{n \rightarrow \infty} u_n = \lim_{n \rightarrow \infty} \frac{2^n - 2}{2^n + 1}$$

$$= \lim_{n \rightarrow \infty} \frac{1 - \frac{2}{2^n}}{1 + \frac{1}{2^n}}$$

$$= 1 \neq 0$$

$\sum u_n$ diverges for $x = 1$. (using zero test)

Hence the given series converges for $x < 1$ & diverges for $x \geq 1$.



When the ratio test fails, we apply the following test:

1) Raabe's test: In the positive term series $\sum u_n$, if

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = k$$

then the series converges for $k > 1$ & diverges for $k < 1$, but the test fails for $k = 1$.

2) Logarithmic test: In the positive term series $\sum u_n$, if

$$\lim_{n \rightarrow \infty} \left(n \log \frac{u_n}{u_{n+1}} \right) = k,$$

then the series converges for $k > 1$ & diverges for $k < 1$, but the test fails for $k = 1$.



$$1) \sum \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \cdot \frac{x^{2n}}{2n} \quad (x > 0)$$

$$u_n = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)}{2 \cdot 4 \cdot 6 \dots 2n} \cdot \frac{x^{2n}}{2n}$$

$$u_{n+1} = \frac{1 \cdot 3 \cdot 5 \dots (2n+1)}{2 \cdot 4 \cdot 6 \dots (2n+2)} \cdot \frac{x^{2n+2}}{2n+2}$$

$$\frac{u_{n+1}}{u_n} = \frac{1 \cdot 3 \cdot 5 \dots (2n-1)(2n+1)}{2 \cdot 4 \cdot 6 \dots 2n(2n+2)} \cdot \frac{x^{2n+2}}{2n+2} \times \frac{2 \cdot 4 \cdot 6 \dots 2n}{1 \cdot 3 \cdot 5 \dots (2n-1)} \cdot \frac{2n}{x^{2n}}$$

$$= \frac{2n+1}{2n+2} \cdot \frac{2n}{2n+2} \cdot x^2$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = x^2$$

By ratio test $\sum u_n$ converges if $x^2 < 1$ i.e., $x < 1$, diverges if $x^2 > 1$, i.e., $x > 1$ & the test fails if $x^2 = 1$



When $x = 1$, we apply Raabe's test

$$\begin{aligned} n \left(\frac{u_n}{u_{n+1}} - 1 \right) &= n \left(\frac{(2n+2)^2}{2n(2n+1)} - 1 \right) \\ &= n \left(\frac{4n^2 + 8n + 4 - 4n^2 - 2n}{2n(2n+1)} \right) \\ &= \frac{3n+2}{2n+1} \end{aligned}$$

$$\lim_{n \rightarrow \infty} n \left(\frac{u_n}{u_{n+1}} - 1 \right) = \frac{3}{2} > 1$$

By Raabe's test, $\sum u_n$ converges when $x = 1$



$$1) 1 + \frac{x}{2} + \frac{2!}{3^2} x^2 + \frac{3!}{4^3} x^3 + \frac{4!}{5^4} x^4 + \dots \quad (x > 0)$$

$$\text{Here } u_n = \frac{n!}{(n+1)^n} x^n \quad u_{n+1} = \frac{(n+1)!}{(n+2)^{n+1}} x^{n+1}$$

$$\begin{aligned} \frac{u_{n+1}}{u_n} &= \frac{(n+1)!}{(n+2)^{n+1}} x^{n+1} \cdot \frac{(n+1)^n}{n! x^n} \\ &= \frac{(n+1)^{n+1}}{(n+2)^{n+1}} x \end{aligned}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} &= \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{1}{n}\right)^n \left(1 + \frac{1}{n}\right)}{\left(1 + \frac{2}{n}\right)^n \left(1 + \frac{1}{n}\right)} x \\ &= \frac{e}{e^2} \cdot x = \frac{x}{e} \end{aligned}$$

By ratio test, $\sum u_n$ converges if $x < e$ and diverges if $x > e$.



The test fails if $x = e$. In this case, the series is

$$1 + \frac{1}{2}e + \frac{2!}{3^2}e^2 + \frac{3!}{4^3}e^3 + \frac{4!}{5^4}e^4 + \dots$$

We apply logarithmic test,

$$\begin{aligned}\log \frac{u_n}{u_{n+1}} &= (n+1) \log \left(1 + \frac{2}{n}\right) - (n+1) \log \left(1 + \frac{1}{n}\right) - \log e \\&= (n+1) \left\{ \left(\frac{2}{n} - \frac{1}{2} \cdot \frac{4}{n^2} + \frac{1}{3} \cdot \frac{8}{n^3} - \frac{1}{4} \cdot \frac{16}{n^4} + \dots \right) \right. \\&\quad \left. - \left(\frac{1}{n} - \frac{1}{2} \cdot \frac{1}{n^2} + \frac{1}{3} \cdot \frac{1}{n^3} + \dots \right) \right\} - 1 \\&= (n+1) \left[\frac{1}{n} - \frac{3}{2n^2} + \frac{7}{3n^3} - \dots \right] - 1 \\&= 1 - \frac{3}{2} \cdot \frac{1}{n} + \frac{7}{3} \cdot \frac{1}{n^2} - \dots + \frac{1}{n} - \frac{3}{2n^2} + \frac{7}{3} \cdot \frac{1}{n^3} - \dots - 1 \\&= -\frac{1}{2} \cdot \frac{1}{n} + \frac{5}{6} \cdot \frac{1}{n^2} - \dots \\(n \log \frac{u_n}{u_{n+1}}) &= -\frac{1}{2} + \frac{5}{6} \cdot \frac{1}{n} - \dots \\ \lim_{n \rightarrow \infty} (n \log \frac{u_n}{u_{n+1}}) &= -\frac{1}{2} < 1\end{aligned}$$

By logarithmic test, the series diverges when $x = e$.